We can now explain the origin of Newton's inverse square law. An imaginary sphere centered on the sun and passing through the satellite's location, as in Figure 13.4c, has a surface area that—like the surface of any sphere in three-dimensional space—is proportional to the *square* of its radius, which in this case is the *square* of the distance between the sun and the satellite. This means that the density of field lines passing through the sphere—the total number of field lines divided by the sphere's area— decreases as the square of sun-satellite separation. If you double the distance, the same number of field lines are now uniformly spread out on a sphere with four times the surface area, and hence the gravitational pull at that distance will drop by a factor of four. Newton's inverse square law for gravity is thus a reflection of a geometrical property of spheres in three space dimensions.

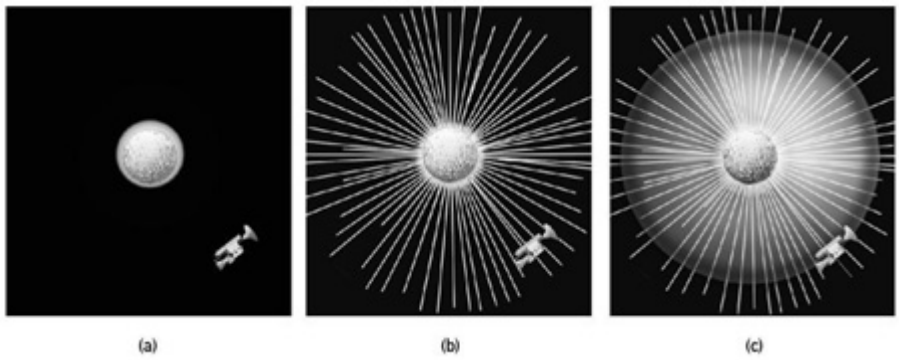


Figure 13.4 **(**a**)** The gravitational force exerted by the sun on an object, such as a satellite, is inversely proportional to the square of the distance between them. The reason is that the sun's gravitational field lines spread out uniformly as in **(**b**)** and hence have a density at a distance *d* that is inversely proportional to the area of an imaginary sphere of radius *d*—schematically drawn in **(**c**)—**an area which basic geometry shows to be proportional to d^2.

By contrast, if the universe had two or even just one space dimension, how would Newton's formula change? Well, Figure 13.5a shows a two-dimensional version of the sun and its orbiting satellite. As you can see, at any given distance the sun's gravitational field lines uniformly spread out on a circle, the analog of a sphere in one lower dimension. Since the circle's circumference is proportional to its radius (not to the square of its radius), if you double the sun–satellite separation, the density of field lines will decrease by a factor of 2 (not 4) and so the strength of the sun's